Codoctor — Model & Data Card

SciBlitz AI Challenge 2026 · Track A. Advisory, non-diagnostic clinical decision-support. Live: https://codoctor.vercel.app · Repo: https://github.com/Nafiz001/codoctor

Intended use & scope

- **Intended use:** an *advisory* co-pilot for a clinician during a short OPD consultation in a low-resource Bangladeshi setting — surfacing cited guideline prompts, a deterministic danger-sign and drug-safety check, and a plain-Bangla spoken take-home summary for the patient.
- **Intended users:** a licensed clinician (or, for the patient summary, the patient) — never an unsupervised layperson for self-treatment.
- **Out of scope:** autonomous diagnosis or prescription; emergencies without a clinician; conditions outside the modelled **pediatric acute respiratory infection (WHO IMCI)** golden path; any use that replaces professional judgement.
- **Human oversight:** every output requires clinician confirmation. The deterministic engines make the high-stakes calls; the LLM only narrates and is forbidden from adding un-grounded claims.

Pre-trained models used

Model / service	Provider	Role	License	Notes
GPT-4o-mini	OpenAI	**Key-optional, two roles:** (1) structured clinical-entity extraction from the fused Bangla transcript, merged *on top of* a deterministic lexicon floor it can add to but never override; (2) Bangla narration — *rephrases already-grounded findings only*, behind a guard rejecting un-grounded drug mentions	Proprietary (commercial API)	**Not required.** System runs fully without any key (regex Scribe + template narration). The LLM never makes a clinical decision. `ANTHROPIC_API_KEY` /Claude Haiku is a supported alternative.
text-embedding-3-small	OpenAI	**Key-optional** dense semantic ranker fused (RRF) with BM25 + TF-IDF for cross-lingual retrieval	Proprietary (commercial API)	Corpus embedded once and cached; only the query embedded per request. Without a key, retrieval is the keyless two-ranker hybrid.
Web Speech API — SpeechRecognition	Browser/OS (e.g. Chrome → Google)	Keyless Bangla speech-to-text on `/room` and `/live`	Browser-provided service	No model shipped or trained by us; availability/quality depend on the user's browser.
Web Speech API — SpeechSynthesis	Browser/OS	Reads the patient's Bangla summary aloud	Browser-provided service	Same as above.
BGE-M3 (`BAAI/bge-m3`)	BAAI	Self-hosted dense retrieval	MIT	**Planned replacement** for the OpenAI embedding dependency; the `retriever.search()` contract is the swap-in point.

No model training or fine-tuning was performed. The deterministic engines and retriever are authored, auditable code.

Data sources

Source	Owner	Use
WHO IMCI chart booklet (cough/difficult-breathing thresholds, danger signs)	World Health Organization	Authored the danger-sign decision tree and cited corpus chunks (paraphrased, with attribution).
DGHS Standard Treatment Guidelines (Bangladesh)	DGHS, Govt. of Bangladesh	Cited corpus chunks for childhood-pneumonia treatment & referral.
National Drug Formulary of Bangladesh / BNF	DGDA / formulary bodies	Drug-class map, contraindication & interaction facts; cited monograph chunks.
Evaluation set	Authored by us	Hand-labelled cases in `backend/eval/run_eval.py` : 17 IMCI classification cases, 13 medication-safety cases, 4 orchestrator cases. Reproduce with `python backend/eval/run_eval.py` .

We reproduce **no copyrighted text verbatim at scale** — corpus entries are concise paraphrases of public clinical guidance, each attributed to its source. No patient data is used; the demo uses synthetic, anonymized cases.

Evaluation results

Software-correctness on our authored set (`python backend/eval/run_eval.py`), **not** patient-outcome accuracy: IMCI classification **17/17**; danger-sign recall **5/5 (0 missed)**; specificity **12/12 (0 false referrals)**; medication-safety catch **7/7** with **0/6 false-positives**; orchestrator grounding + citation **3/3**; honest refusal on insufficient data **1/1**. Unit tests **29/29** — all measured on the keyless deterministic path.

Known limitations & ethical considerations

- **Advisory & non-diagnostic.** Does not diagnose, prescribe, or replace a clinician; every output requires clinician confirmation. Deterministic engines make the high-stakes calls; the LLM only narrates.
- **Not clinically validated.** Reported numbers are software-correctness on a curated set, **not** patient-outcome accuracy. Scope is currently **pediatric acute respiratory infection (WHO IMCI)** only; a small corpus over-represents the modelled condition.
- **Guideline currency.** Paraphrased chunks may lag official updates — always defer to the current WHO/DGHS/NDF source.
- **Voice reliability.** Browser Bangla ASR degrades on dialect/noise and is unavailable on iOS Safari; mitigated by dual-mic fusion, a typed/seeded fallback, and the scripted `/doctor` path. No speaker diarization — streams are fused by timing + lexical overlap (genuine same-utterance merges only).

- **Engineering limits (honest).** Live session is in-memory (single process, ~2h TTL; a restart clears it) and syncs by ~3s HTTP polling, not WebSockets; the patient summary persists only on the patient's device. No durable longitudinal record, dense retriever, or clinician co-sign yet.
- **Privacy & misuse.** Processed in-session; the patient owns their record; no PII retained server-side. Unsupervised self-treatment is an explicit risk — the UI keeps a clinician/facility in the loop and states the advisory scope. Broadening coverage and clinical co-sign are required before any real-world use.